

DBMS LAB

ER Model and ER-to-Relational Mapping

Assignment - 2

Submitted by: Ganji Satwika

Roll No: 24MAB0A09

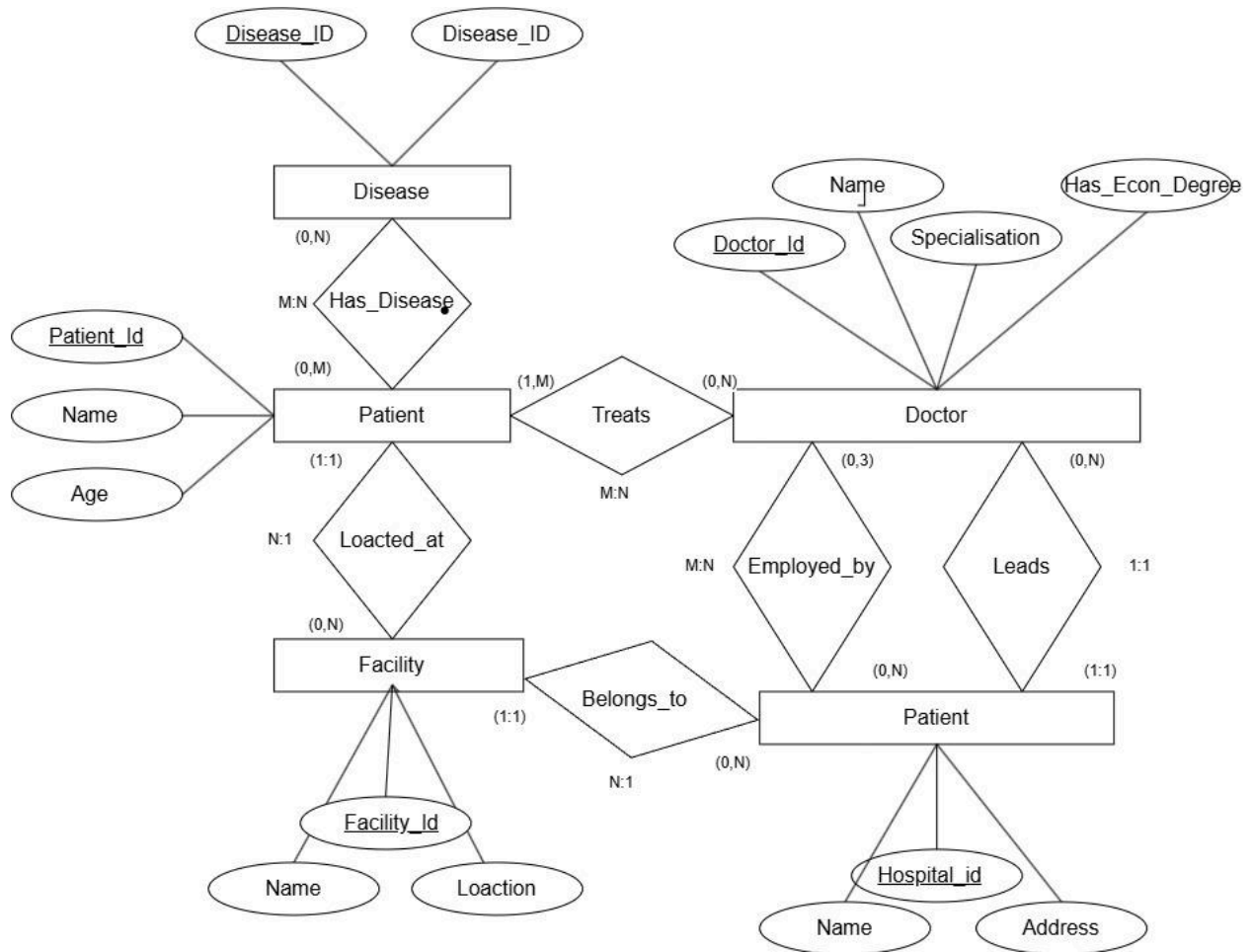
Submitted to: Dr. Vikas Srivastava

Course: DBMS Lab

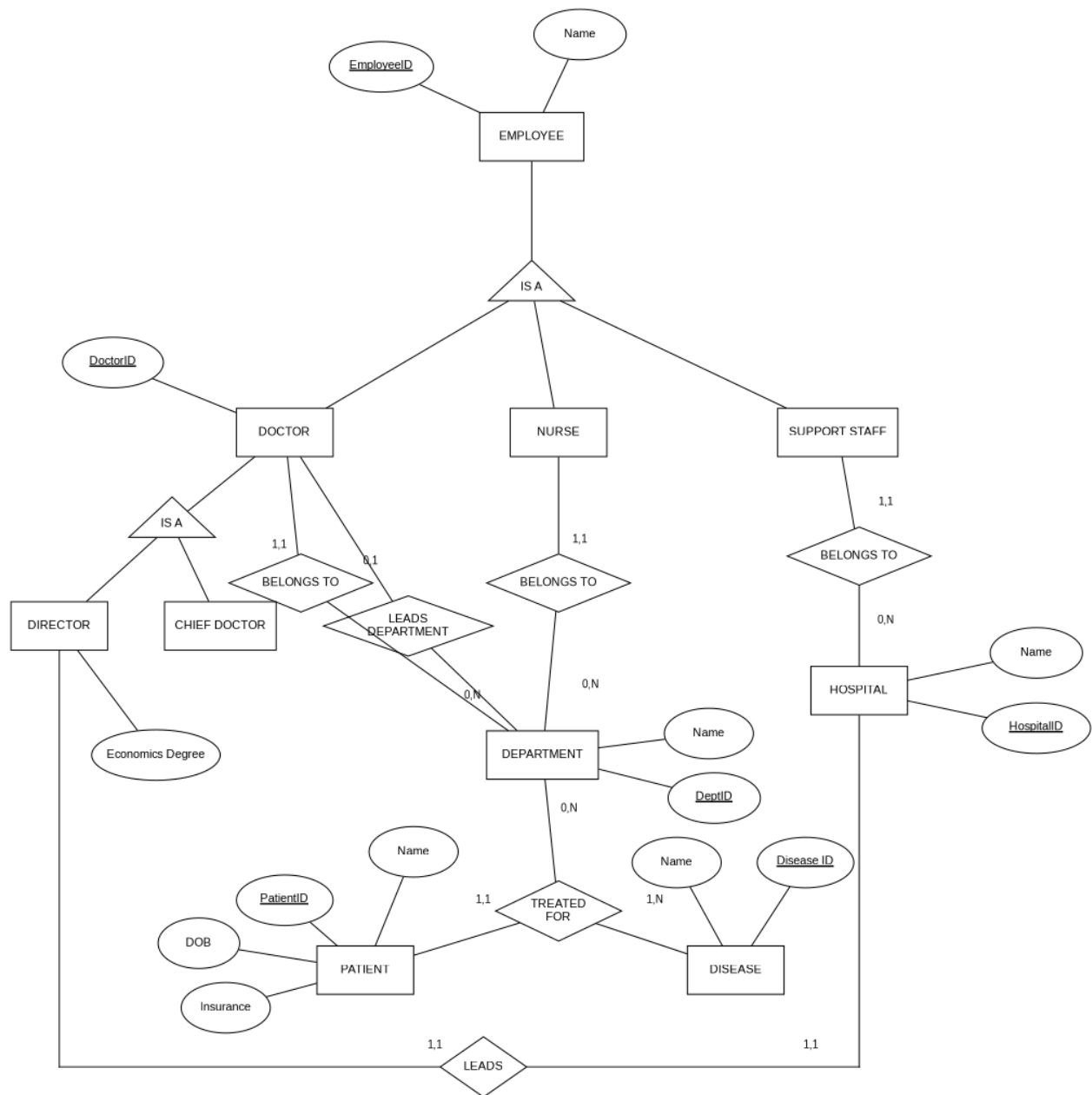
Date: 26 August 2026

Question 1

ER Diagram:



Question 2



Question 3

Since every relationship (P, Q, R, T, S, U) has exactly one arrowhead, each one is many-to-one. A many-to-one relationship never needs its own table — it's represented by adding a foreign key to the entity on the "many" side. So we end up with exactly 4 relations (one per entity), which is the minimum possible.

R1: A(A1, A2, B1)

- Primary Key: A1
- Foreign Key: B1 $\rightarrow$ B.B1
- B1 is included because relationship P is many-to-one from A to B (many A's relate to one B), so A gets the foreign key.

R2: B(B1, B2, C1)

- Primary Key: B1
- Foreign Key: C1 $\rightarrow$ C.C1
- C1 is included because relationship Q is many-to-one from B to C, so B gets the foreign key.

R3: C(C1, C2, D1)

- Primary Key: C1
- Foreign Key: D1 $\rightarrow$ D.D1
- D1 is included because relationship R is many-to-one from C to D, so C gets the foreign key.

R4: D(D1, D2, D3, A1_T, A1_S, A1_U)

- Primary Key: D1
- Foreign Keys: A1_T $\rightarrow$ A.A1, A1_S $\rightarrow$ A.A1, A1_U $\rightarrow$ A.A1
- T, S, and U are three separate many-to-one relationships from D to A, so D needs three distinct foreign key columns (one per relationship) — a single column can't do the job of all three at once.

Total: 4 relations, since none of the six relationships is many-to-many, so none required a standalone relationship table.

Question 4

(a) Transformation of the ER diagrams into relational schemas

1. Relational schemas for Question 1

The ER model of Question 1 contains the entity sets Patient, Disease, Facility, Hospital, and Doctor. The relationships are represented using foreign keys and separate relations where necessary.

Hospital

HOSPITAL(Hospital_ID, Hospital_Name, Director_Doctor_ID)

- Primary Key: Hospital_ID
- Foreign Key: Director_Doctor_ID $\rightarrow$ DOCTOR(Doctor_ID)

Doctor

DOCTOR(Doctor_ID, Doctor_Name)

- Primary Key: Doctor_ID

Facility

FACILITY(Facility_ID, Facility_Name, Hospital_ID)

- Primary Key: Facility_ID
- Foreign Key: Hospital_ID $\rightarrow$ HOSPITAL(Hospital_ID)

Each facility belongs to exactly one hospital.

Patient

PATIENT(Patient_ID, Patient_Name, Facility_ID)

- Primary Key: Patient_ID
- Foreign Key: Facility_ID $\rightarrow$ FACILITY(Facility_ID)

Each patient is treated at exactly one facility.

Disease

DISEASE(Disease_ID, Disease_Name)

- Primary Key: Disease_ID

Patient–Disease

Since a patient may have multiple diseases and a disease may occur in multiple patients, this is a many-to-many relationship.

PATIENT_DISEASE(Patient_ID, Disease_ID)

- Primary Key: (Patient_ID, Disease_ID)
- Foreign Key: Patient_ID → PATIENT(Patient_ID)
- Foreign Key: Disease_ID → DISEASE(Disease_ID)

Doctor–Patient

A doctor may treat multiple patients and a patient may be treated by multiple doctors.

DOCTOR_PATIENT(Doctor_ID, Patient_ID)

- Primary Key: (Doctor_ID, Patient_ID)
- Foreign Key: Doctor_ID → DOCTOR(Doctor_ID)
- Foreign Key: Patient_ID → PATIENT(Patient_ID)

Doctor–Hospital Employment

A doctor can be employed by a maximum of three hospitals, while a hospital can employ multiple doctors.

DOCTOR_HOSPITAL(Doctor_ID, Hospital_ID)

- Primary Key: (Doctor_ID, Hospital_ID)
- Foreign Key: Doctor_ID → DOCTOR(Doctor_ID)
- Foreign Key: Hospital_ID → HOSPITAL(Hospital_ID)

A constraint must ensure that each doctor is associated with no more than three hospitals.

Hospital Director

The director of a hospital is a doctor of that hospital, has an economics degree, and does not work in any other hospital. This can be represented by Director_Doctor_ID in HOSPITAL.

The following constraints apply:

- Director_Doctor_ID must reference DOCTOR(Doctor_ID).
- The director must also occur in DOCTOR_HOSPITAL for the same hospital.
- A director cannot be employed by another hospital.
- A doctor can be the director of at most one hospital.

Relational schemas for Question 2

The ER model of Question 2 contains Hospital, Department, Employee and its specialized employee types, Patient, and Disease. Doctors and nurses are assigned to departments, while support staff can belong directly to the hospital.

Hospital

HOSPITAL(Hospital_ID, Hospital_Name, Director_Doctor_ID)

- Primary Key: Hospital_ID
- Foreign Key: Director_Doctor_ID $\rightarrow$ DOCTOR(Employee_ID)

The director must be a doctor employed by the same hospital and must not be employed by another hospital.

Employee

EMPLOYEE(Employee_ID, Employee_Name, Hospital_ID, Medical_Degree)

- Primary Key: Employee_ID
- Foreign Key: Hospital_ID $\rightarrow$ HOSPITAL(Hospital_ID)

Each employee belongs to exactly one hospital.

Doctor

DOCTOR(Employee_ID, Doctors_Association_ID, Department_ID)

- Primary Key: Employee_ID
- Foreign Key: Employee_ID $\rightarrow$ EMPLOYEE(Employee_ID)
- Foreign Key: Department_ID $\rightarrow$ DEPARTMENT(Department_ID)

Nurse

NURSE(Employee_ID, Department_ID)

- Primary Key: Employee_ID
- Foreign Key: Employee_ID $\rightarrow$ EMPLOYEE(Employee_ID)
- Foreign Key: Department_ID $\rightarrow$ DEPARTMENT(Department_ID)

Support Staff

SUPPORT_STAFF(Employee_ID)

- Primary Key: Employee_ID
- Foreign Key: Employee_ID $\rightarrow$ EMPLOYEE(Employee_ID)

Support staff may belong directly to the hospital and therefore do not require a Department_ID.

Chief Doctor

A chief doctor is a specialized doctor.

CHIEF_DOCTOR(Employee_ID)

- Primary Key: Employee_ID
- Foreign Key: Employee_ID → DOCTOR(Employee_ID)

Department

DEPARTMENT(Department_ID, Department_Name, Hospital_ID, Leader_Employee_ID, Commissioned_Leader_ID)

- Primary Key: Department_ID
- Foreign Key: Hospital_ID → HOSPITAL(Hospital_ID)
- Foreign Key: Leader_Employee_ID → CHIEF_DOCTOR(Employee_ID)
- Foreign Key: Commissioned_Leader_ID → EMPLOYEE(Employee_ID)

A department belongs to exactly one hospital.

If a department has a chief-doctor leader, Leader_Employee_ID identifies that chief doctor. If there is no chief-doctor leader, Commissioned_Leader_ID identifies the commissioned department leader.

Patient

PATIENT(Patient_ID, Patient_Name)

- Primary Key: Patient_ID

Disease

DISEASE(Disease_ID, Disease_Name)

- Primary Key: Disease_ID

Patient–Department Treatment

A patient may be treated in multiple departments, and a department may treat multiple patients.

PATIENT_DEPARTMENT(Patient_ID, Department_ID)

- Primary Key: (Patient_ID, Department_ID)
- Foreign Key: Patient_ID → PATIENT(Patient_ID)
- Foreign Key: Department_ID → DEPARTMENT(Department_ID)

Patient–Disease Treatment

A patient may be treated for different diseases during the hospital stay, and a disease may be treated in relation to many patients.

PATIENT_DISEASE(Patient_ID, Disease_ID)

- Primary Key: (Patient_ID, Disease_ID)
- Foreign Key: Patient_ID → PATIENT(Patient_ID)
- Foreign Key: Disease_ID → DISEASE(Disease_ID)

(b) Transformation using as few separate relational schemas as possible

For one-to-many relationships, the foreign key can be placed in the relation on the many side. Therefore, separate relations are not required for those relationships. Many-to-many relationships still require separate relations.

Minimum-schema transformation for Question 1

HOSPITAL(Hospital_ID, Hospital_Name, Director_Doctor_ID)

- Primary Key: Hospital_ID
- Foreign Key: Director_Doctor_ID → DOCTOR(Doctor_ID)

DOCTOR(Doctor_ID, Doctor_Name)

- Primary Key: Doctor_ID

FACILITY(Facility_ID, Facility_Name, Hospital_ID)

- Primary Key: Facility_ID
- Foreign Key: Hospital_ID → HOSPITAL(Hospital_ID)

PATIENT(Patient_ID, Patient_Name, Facility_ID)

- Primary Key: Patient_ID
- Foreign Key: Facility_ID → FACILITY(Facility_ID)

DISEASE(Disease_ID, Disease_Name)

- Primary Key: Disease_ID

PATIENT_DISEASE(Patient_ID, Disease_ID)

- Primary Key: (Patient_ID, Disease_ID)

- Foreign Keys:
 - Patient_ID → PATIENT(Patient_ID)
 - Disease_ID → DISEASE(Disease_ID)

DOCTOR_PATIENT(Doctor_ID, Patient_ID)

- Primary Key: (Doctor_ID, Patient_ID)
- Foreign Keys:
 - Doctor_ID → DOCTOR(Doctor_ID)
 - Patient_ID → PATIENT(Patient_ID)

DOCTOR_HOSPITAL(Doctor_ID, Hospital_ID)

- Primary Key: (Doctor_ID, Hospital_ID)
- Foreign Keys:
 - Doctor_ID → DOCTOR(Doctor_ID)
 - Hospital_ID → HOSPITAL(Hospital_ID)

Thus, the one-to-many relationships Facility–Hospital and Patient–Facility are represented using foreign keys rather than separate relationship relations.

Minimum-schema transformation for Question 2

HOSPITAL(Hospital_ID, Hospital_Name, Director_Doctor_ID)

- Primary Key: Hospital_ID
- Foreign Key: Director_Doctor_ID → DOCTOR(Employee_ID)

EMPLOYEE(Employee_ID, Employee_Name, Hospital_ID, Medical_Degree)

- Primary Key: Employee_ID
- Foreign Key: Hospital_ID → HOSPITAL(Hospital_ID)

DEPARTMENT(Department_ID, Department_Name, Hospital_ID, Leader_Employee_ID, Commissioned_Leader_ID)

- Primary Key: Department_ID
- Foreign Keys:
 - Hospital_ID → HOSPITAL(Hospital_ID)
 - Leader_Employee_ID → CHIEF_DOCTOR(Employee_ID)
 - Commissioned_Leader_ID → EMPLOYEE(Employee_ID)

DOCTOR(Employee_ID, Doctors_Association_ID, Department_ID)

- Primary Key: Employee_ID
- Foreign Keys:

- Employee_ID → EMPLOYEE(Employee_ID)
- Department_ID → DEPARTMENT(Department_ID)

NURSE(Employee_ID, Department_ID)

- Primary Key: Employee_ID
- Foreign Keys:
 - Employee_ID → EMPLOYEE(Employee_ID)
 - Department_ID → DEPARTMENT(Department_ID)

SUPPORT_STAFF(Employee_ID)

- Primary Key: Employee_ID
- Foreign Key: Employee_ID → EMPLOYEE(Employee_ID)

CHIEF_DOCTOR(Employee_ID)

- Primary Key: Employee_ID
- Foreign Key: Employee_ID → DOCTOR(Employee_ID)

PATIENT(Patient_ID, Patient_Name)

- Primary Key: Patient_ID

DISEASE(Disease_ID, Disease_Name)

- Primary Key: Disease_ID

PATIENT_DEPARTMENT(Patient_ID, Department_ID)

- Primary Key: (Patient_ID, Department_ID)
- Foreign Keys:
 - Patient_ID → PATIENT(Patient_ID)
 - Department_ID → DEPARTMENT(Department_ID)

PATIENT_DISEASE(Patient_ID, Disease_ID)

- Primary Key: (Patient_ID, Disease_ID)
- Foreign Keys:
 - Patient_ID → PATIENT(Patient_ID)
 - Disease_ID → DISEASE(Disease_ID)

The one-to-many relationships are incorporated directly into the corresponding relations using foreign keys, while the many-to-many relationships are represented by separate relational schemas.

Question 5

The ER diagram contains four entity sets E1, E2, E3, and E4, with relationships R1–R5. The original attributes do not allow NULL values.

Relational Schemas

E1

E1(A11, A12, A13)

- Primary Key: A11
- A11: NOT NULL
- A12: NOT NULL
- A13: NOT NULL

E2

E2(A21, A22, E1_A11)

- Primary Key: A21
- A21: NOT NULL
- A22: NOT NULL
- E1_A11: NOT NULL
- Foreign Key: E1_A11 → E1(A11)

R1 is a 1:N relationship, so the foreign key is placed in E2. Since every E2 participates in R1, the foreign key cannot be NULL.

E3

E3(A31, E1_R4_A11, E1_R5_A11, AR4)

- Primary Key: A31
- A31: NOT NULL
- E1_R4_A11: NULL allowed
- E1_R5_A11: NULL allowed
- AR4: NULL allowed
- Foreign Keys:
 - E1_R4_A11 → E1(A11)
 - E1_R5_A11 → E1(A11)

R4 and R5 are both 1:N relationships with E3 on the N-side. Since E3 has optional participation, the corresponding foreign keys may be NULL. AR4 may also be NULL because R4 is optional.

E4

E4(A41, E2_A21, E3_A31)

- Primary Key: A41
- A41: NOT NULL
- E2_A21: NOT NULL
- E3_A31: NOT NULL
- Foreign Keys:
 - E2_A21 $\rightarrow$ E2(A21)
 - E3_A31 $\rightarrow$ E3(A31)

R2 and R3 are 1:N relationships, with E4 on the N-side. Every E4 participates in both relationships, so both foreign keys are NOT NULL.

Final Relational Schemas

E1(A11, A12, A13)

PK: A11

E2(A21, A22, E1_A11)

PK: A21

FK: E1_A11 $\rightarrow$ E1(A11)

E1_A11: NOT NULL

E3(A31, E1_R4_A11, E1_R5_A11, AR4)

PK: A31

FKs: E1_R4_A11 $\rightarrow$ E1(A11), E1_R5_A11 $\rightarrow$ E1(A11)

E1_R4_A11, E1_R5_A11, AR4: NULL allowed

E4(A41, E2_A21, E3_A31)

PK: A41

FKs: E2_A21 $\rightarrow$ E2(A21), E3_A31 $\rightarrow$ E3(A31)

Both foreign keys: NOT NULL